The purpose of this document is to brief the candidates on an interview task

Backend Engineer Assignment (PHP / Laravel)

Objective

Build a High-Traffic News Feed API similar to Twitter / Instagram feed. The goal is to test how the candidate designs efficient queries, handles large datasets, and thinks about scalability.

1. Users

Users can register and follow other users. Users table fields: id, name, email, password.

2. Posts

Users can create posts. Posts table fields: id, user_id, content, created_at.

3. Feed API (Important)

Return posts from followed users. Requirements: latest posts from followed users, pagination support, sorted by newest first, and avoid N+1 query problems.

4. Likes

Users can like posts. API should return total likes and whether the user liked the post.

5. Performance Challenge

Assume the system has 5 million users and 500k posts per day. In the README explain how you would scale feed generation, avoid slow queries, which indexes to add, and when caching should be used.

6. Database Optimization

Design proper indexes, avoid heavy joins when possible, and prevent N+1 query issues.

7. Queue System

When a post is created, send notifications to followers using Laravel Queue. Explain the queue design in the README.

8. Caching

Use caching for feed responses and like counts. Example tools: Redis or Memcached.

9. Code Quality

Follow Laravel best practices, clean controllers, service or repository pattern, proper validation, and proper error handling.

Bonus (Optional)

Docker setup, unit tests, API documentation, or a feed caching strategy.

Submission

Provide a GitHub repository and README including setup instructions, architecture decisions, and scaling strategy.

User Endpoints

Method	Endpoint	Description
POST	/register	Create user
POST	/login	Login
POST	/follow/{user_id}	Follow user
DELETE	/follow/{user_id}	Unfollow user

Post Endpoints

Method	Endpoint	Description
POST	/posts	Create post
GET	/posts/{id}	View post
DELETE	/posts/{id}	Delete post

Like Endpoints

Method	Endpoint
POST	/posts/{id}/like
DELETE	/posts/{id}/like

Augustus Overview

Augustus is a Middle Eastern digital media holding company based in Dubai, Cairo and Riyadh and has been designed to adapt and evolve to this perpetually changing media environment.

We apply our expertise to building media brands, each centered on an audience first mindset, such as promoting the ideals of a positive lifestyle or an aspirational business mentality. Our brands have originated from new media platforms and power our entire ecosystem of services, especially creating brand communications that resonate with our audiences across digital and the real world. Our content is loved and seen by millions, natively across visual, audio and text platforms.

Lovin'

Lovin' is a local news and lifestyle brand. Lovin' captures the best of the world around us: what we see, do, think, eat and drink. Lovin' presents a positive view of our city, town, country or community. Lovin's mantra is simple: Lovin' Your Life.

Smashi

Smashi is a business news network streaming live from the heart of the Arab world, covering the most innovative products, services and technologies changing our lives. Smashi is a mindset, it's for the driven, the dreamers and the doers.

ODEUM

ODEUM is our in-house content studio, powered by data and insights, designed to produce new media formats, in real time for clients that live across our brand channels. We connect brands with communities and culture.